

MDK Arm

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MDK Arm (Microcontroller Development Kit for ARM): An integrated development environment (IDE) for microcontroller applications based on Cortex-M, Cortex-R4, ARM7, and ARM9 processors.

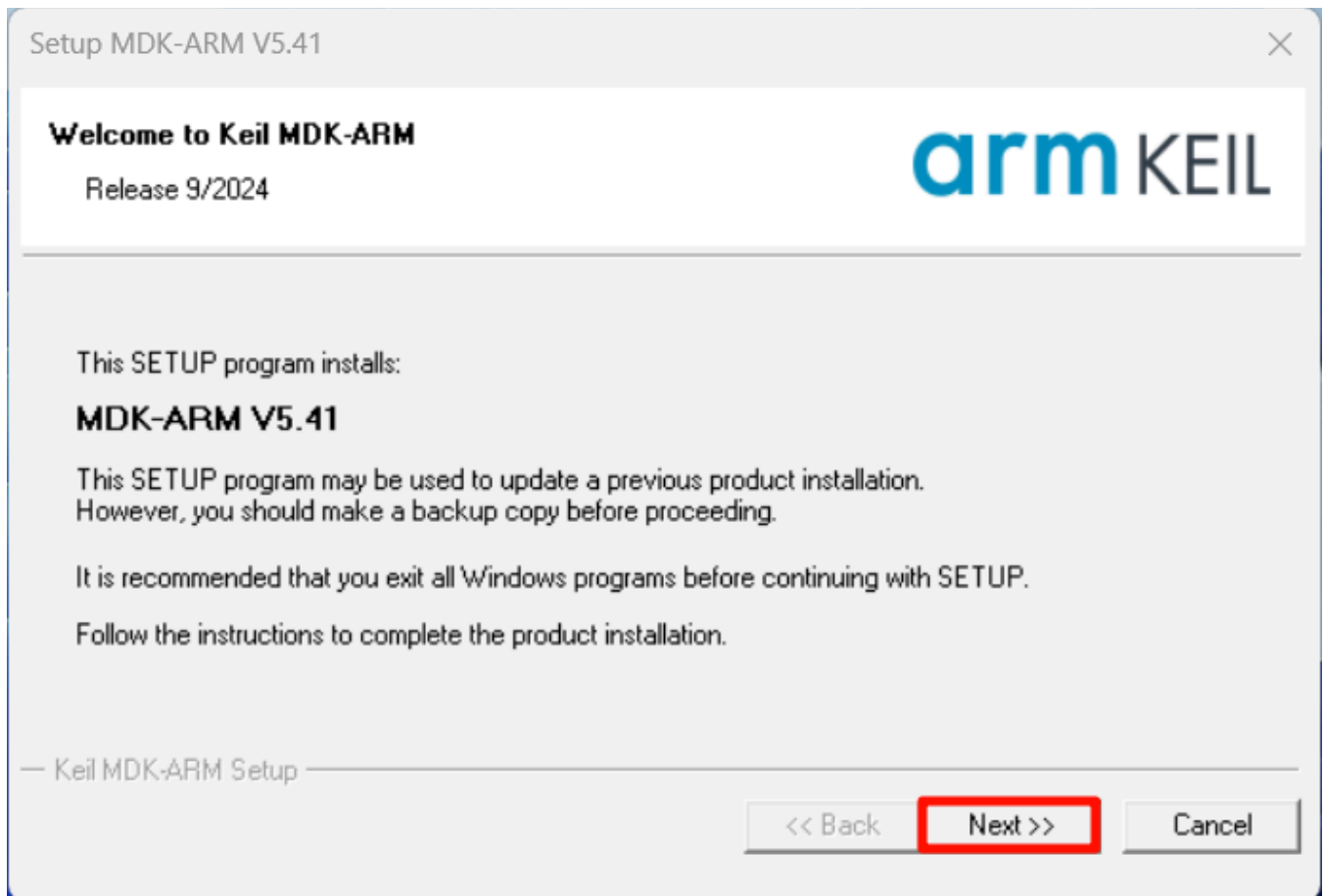
The installation package path and software installation path should not contain Chinese characters. Please place the installation package in a fully English path for installation.

1. MDK-ARM Installation

Building the TI development environment requires Keil MDK-Arm uVision v5.38a or higher!

Since MDK-ARM is commercial software, this tutorial only covers the MDK-ARM installation process.

Open `MDK_541.exe` as administrator and follow the prompts:

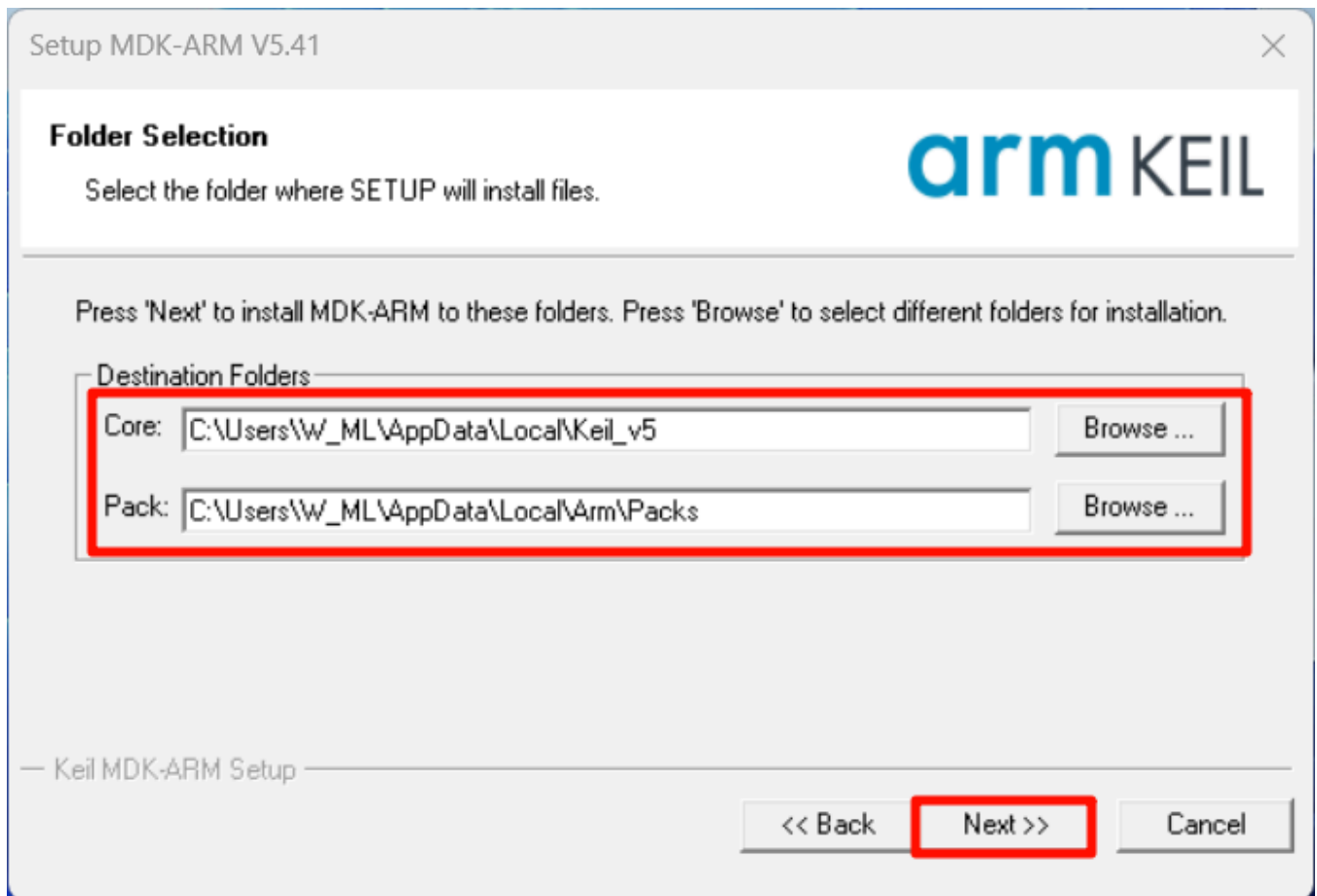


Accept Agreement



Installation Location

It is recommended to use the default location:



User Information

You can fill in arbitrarily:

Setup MDK-ARM V5.41

Customer Information

Please enter your information.

Please enter your name, the name of the company for whom you work and your E-mail address.

First Name: [01]

Last Name: [01]

Company Name: [01]

E-mail: [01]

— Keil MDK-ARM Setup —

<< Back **Next >>** Cancel

If a driver installation popup appears during installation, select Install for all:

Setup MDK-ARM V5.41

Keil MDK-ARM Setup completed

MDK-ARM V5.41

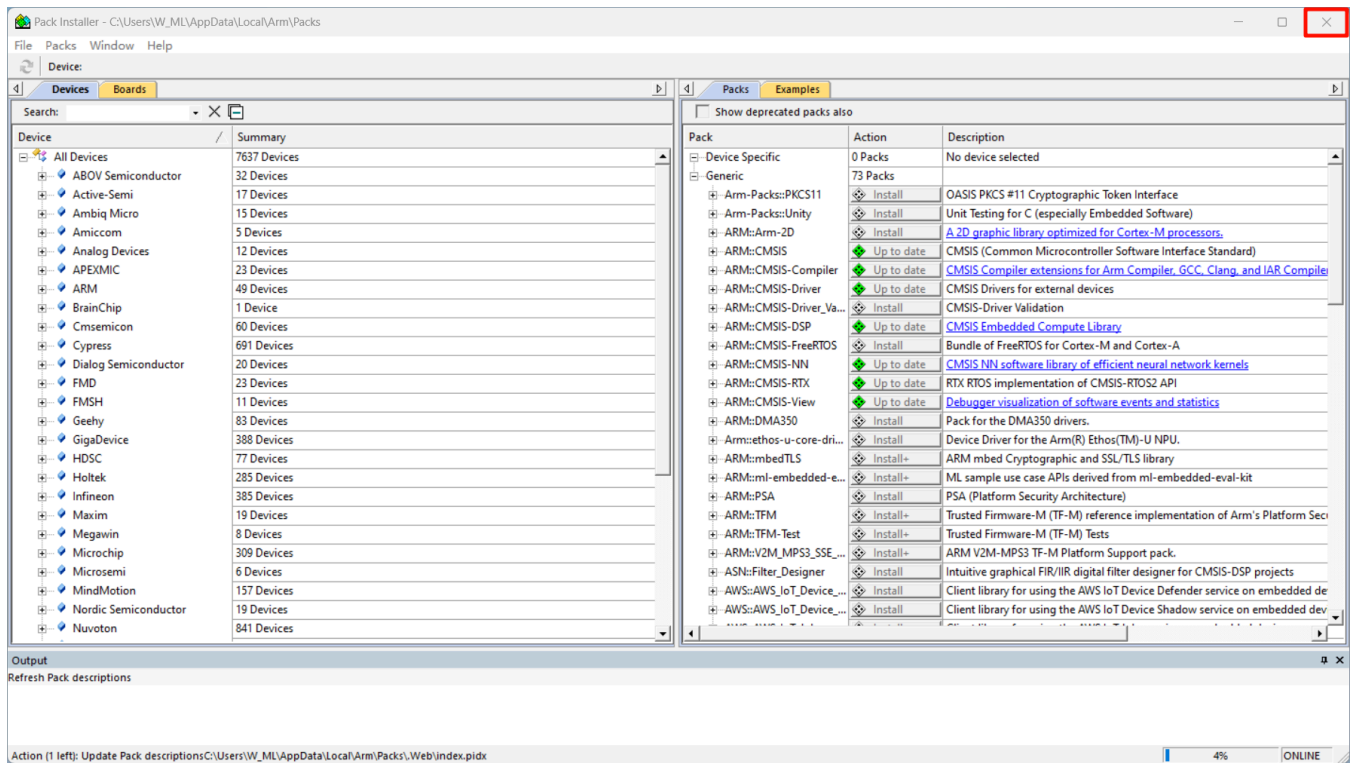
MDK-ARM Core Setup has performed all requested operations successfully.

☐ Show Release Notes.

— Keil MDK-ARM Setup —

<< Back **Finish** Cancel

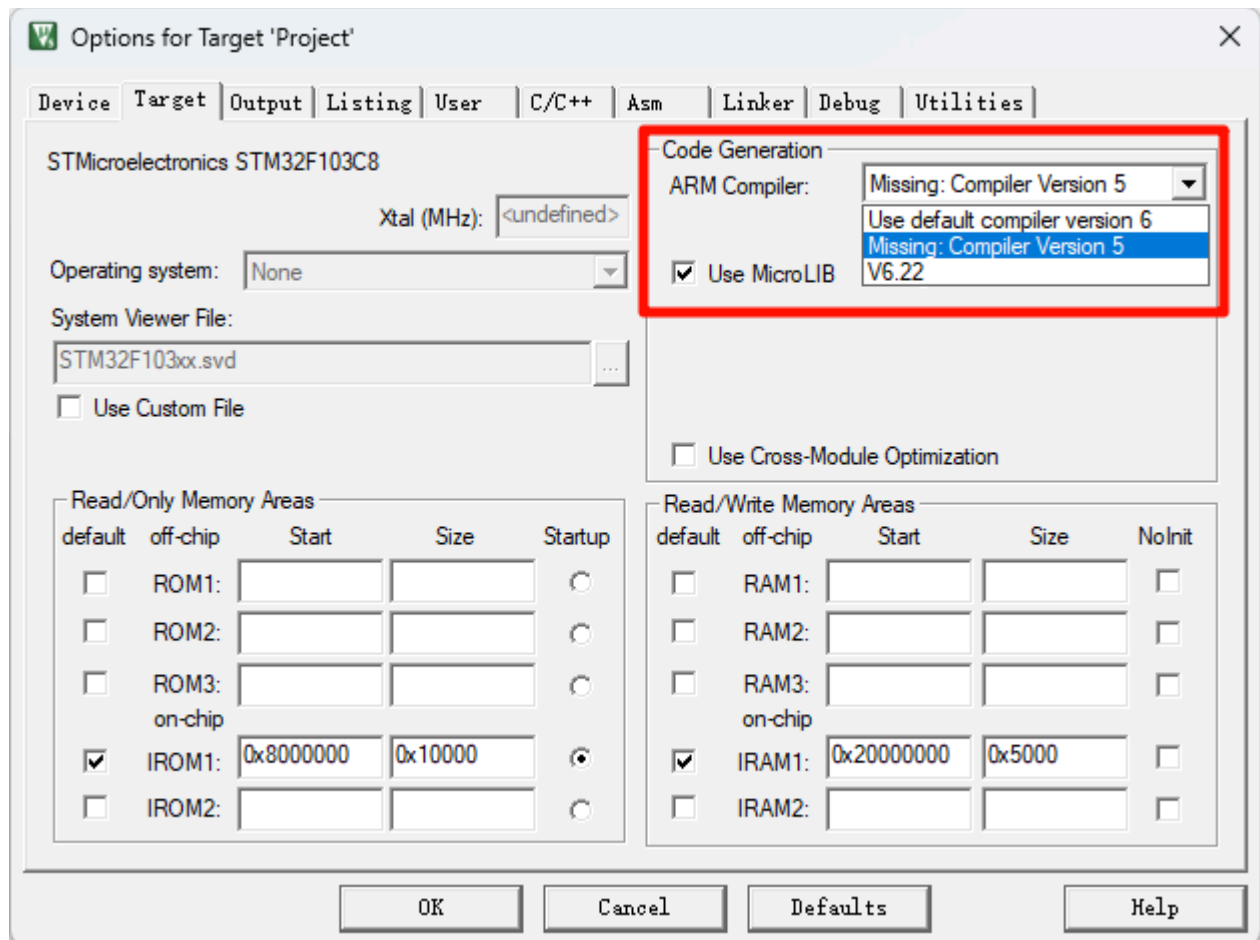
When opening for the first time, you need to close the software package/component interface: This interface can install chip packs but has network requirements!



ARM Compiler 5 (AC 5)

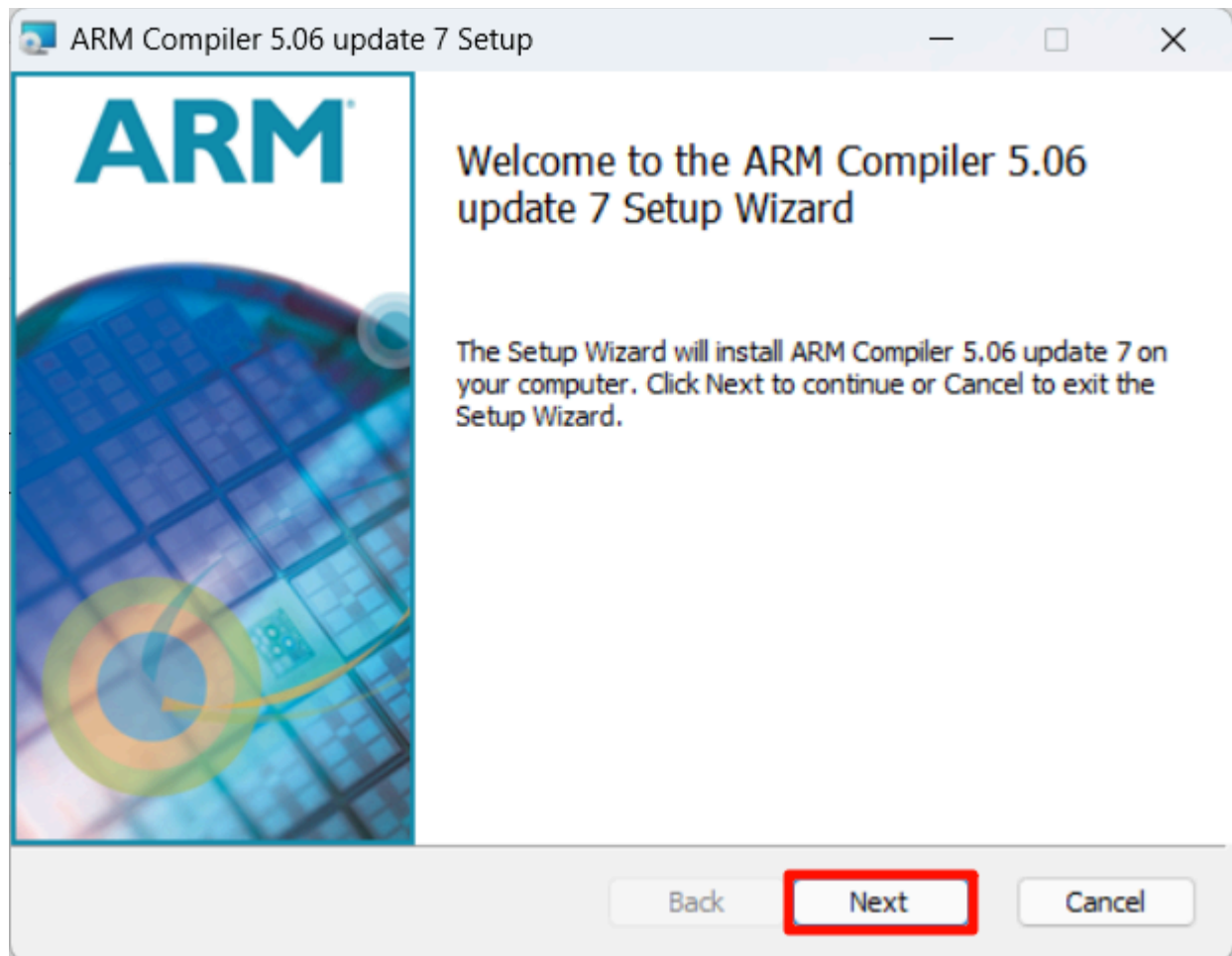
Starting from MDK version 5.37, the AC5 compiler is no longer installed by default and needs to be installed separately!

Using older version project source code for compilation requires AC5. To install AC5 compiler, you need to first open a project.

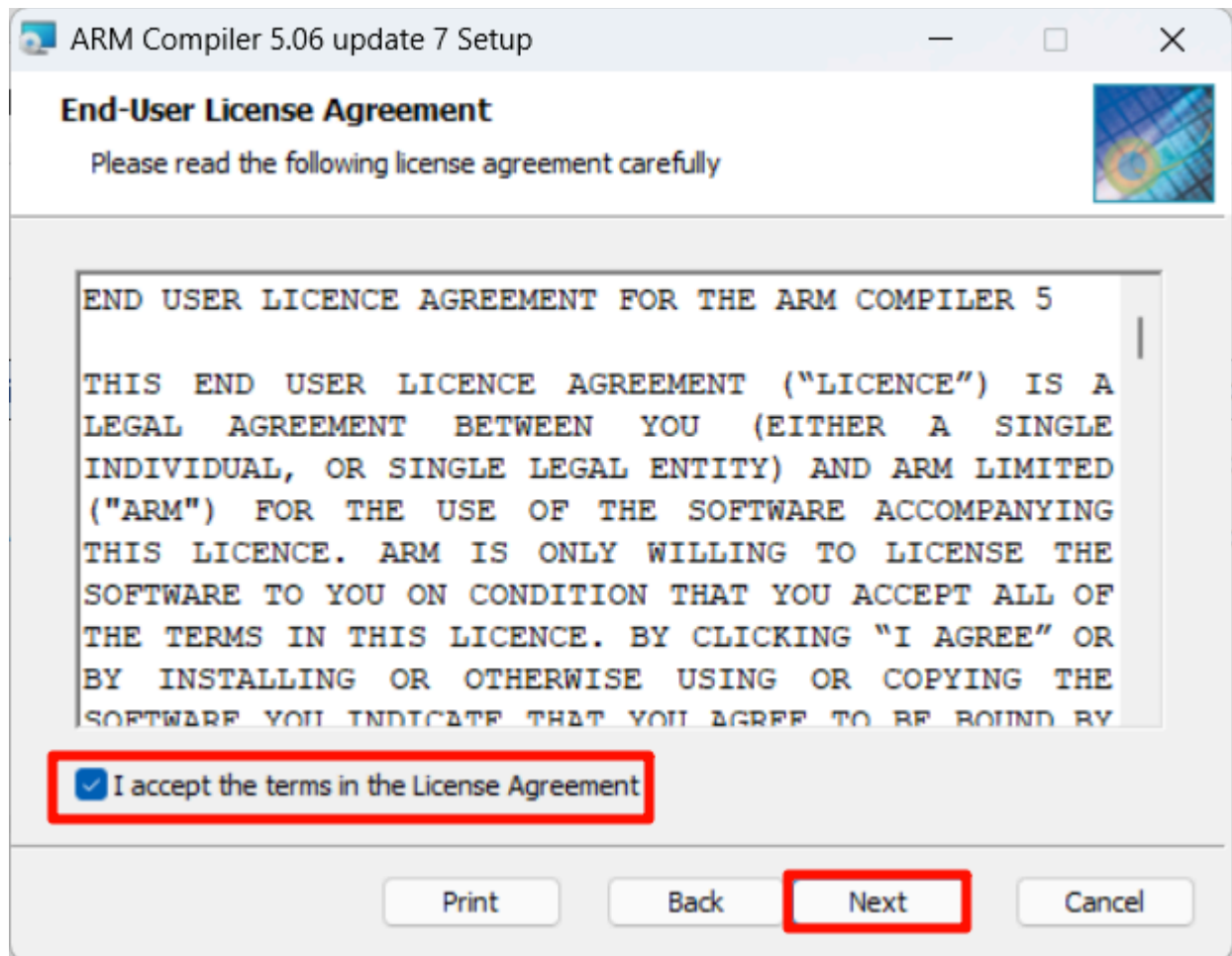


Install AC 5

Open `setup.exe` under `ARMCompiler_5.zip` as administrator and follow the prompts:



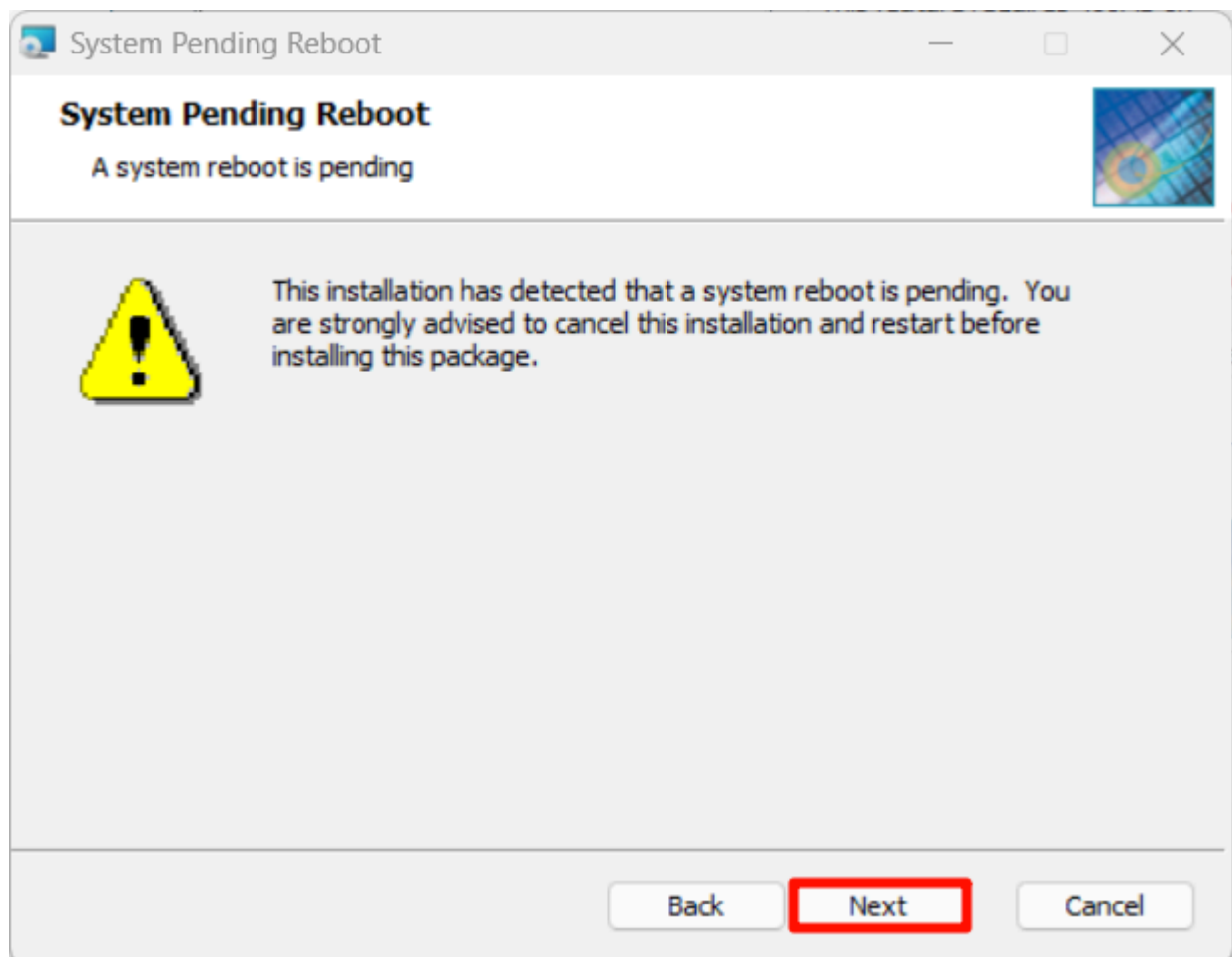
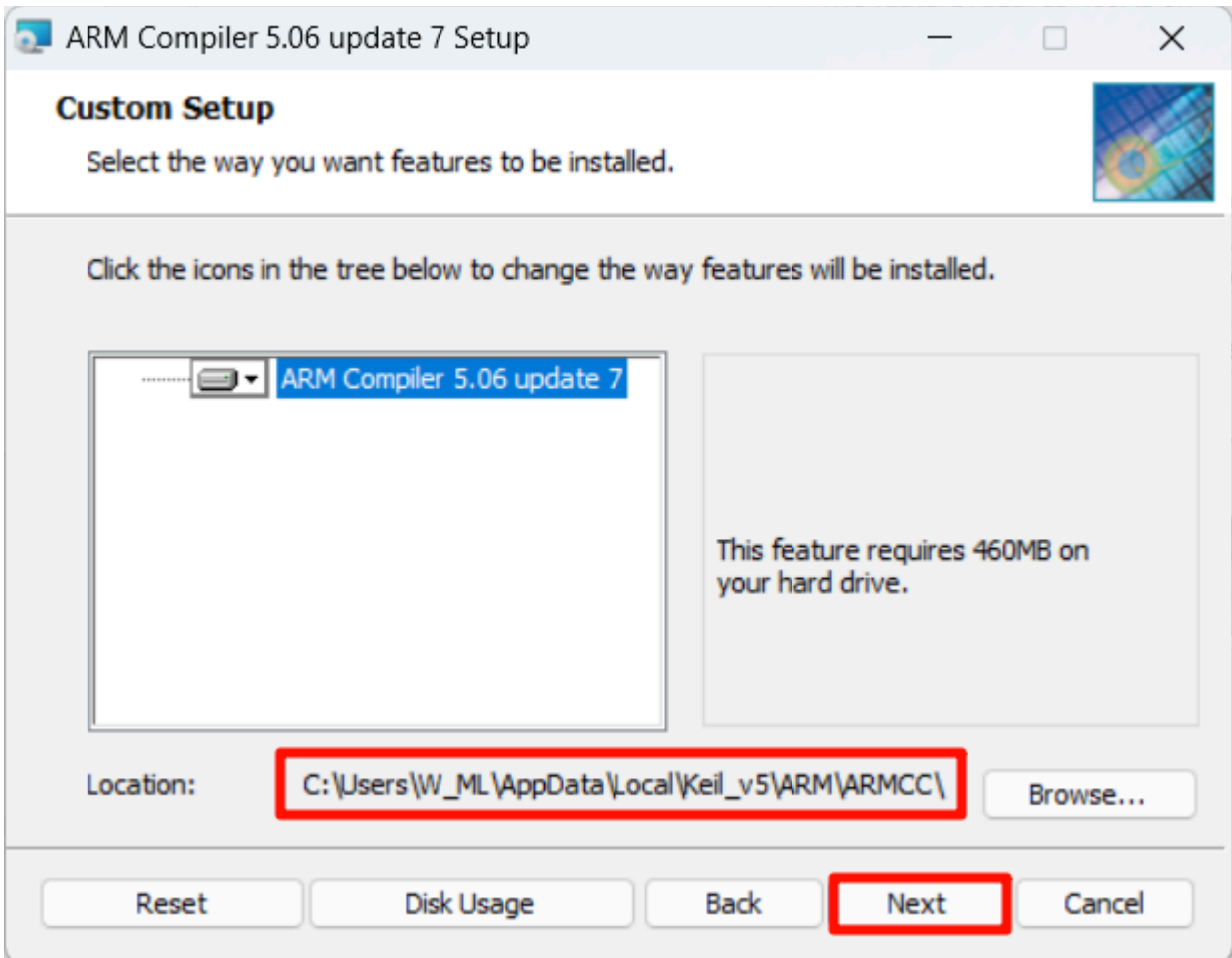
Accept Agreement

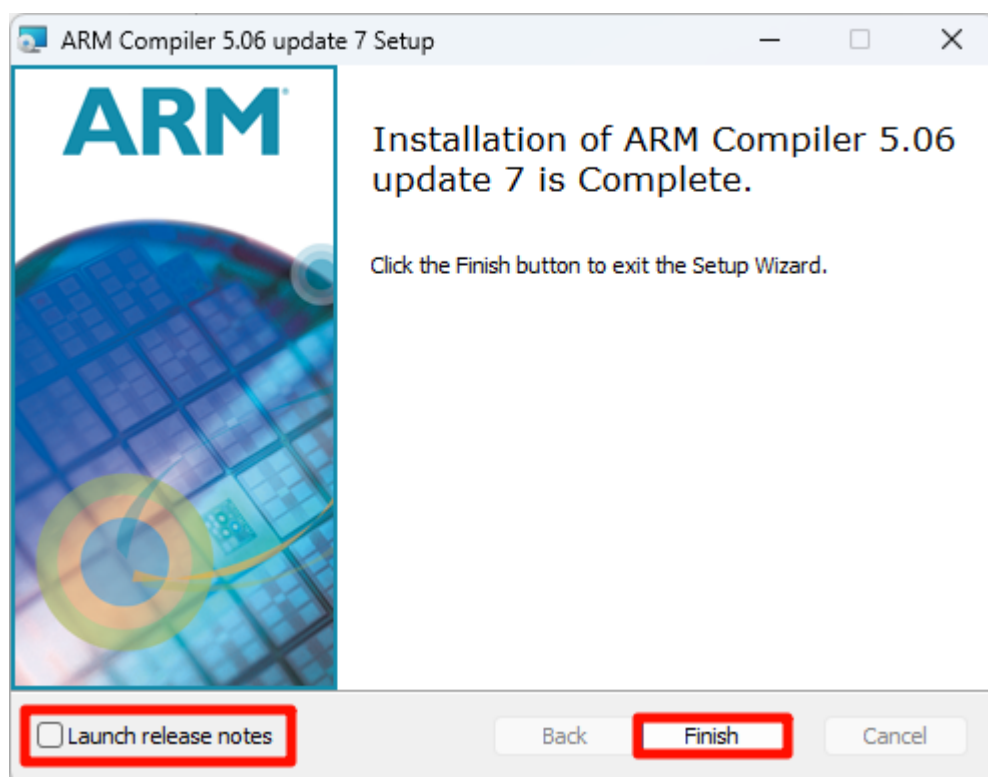
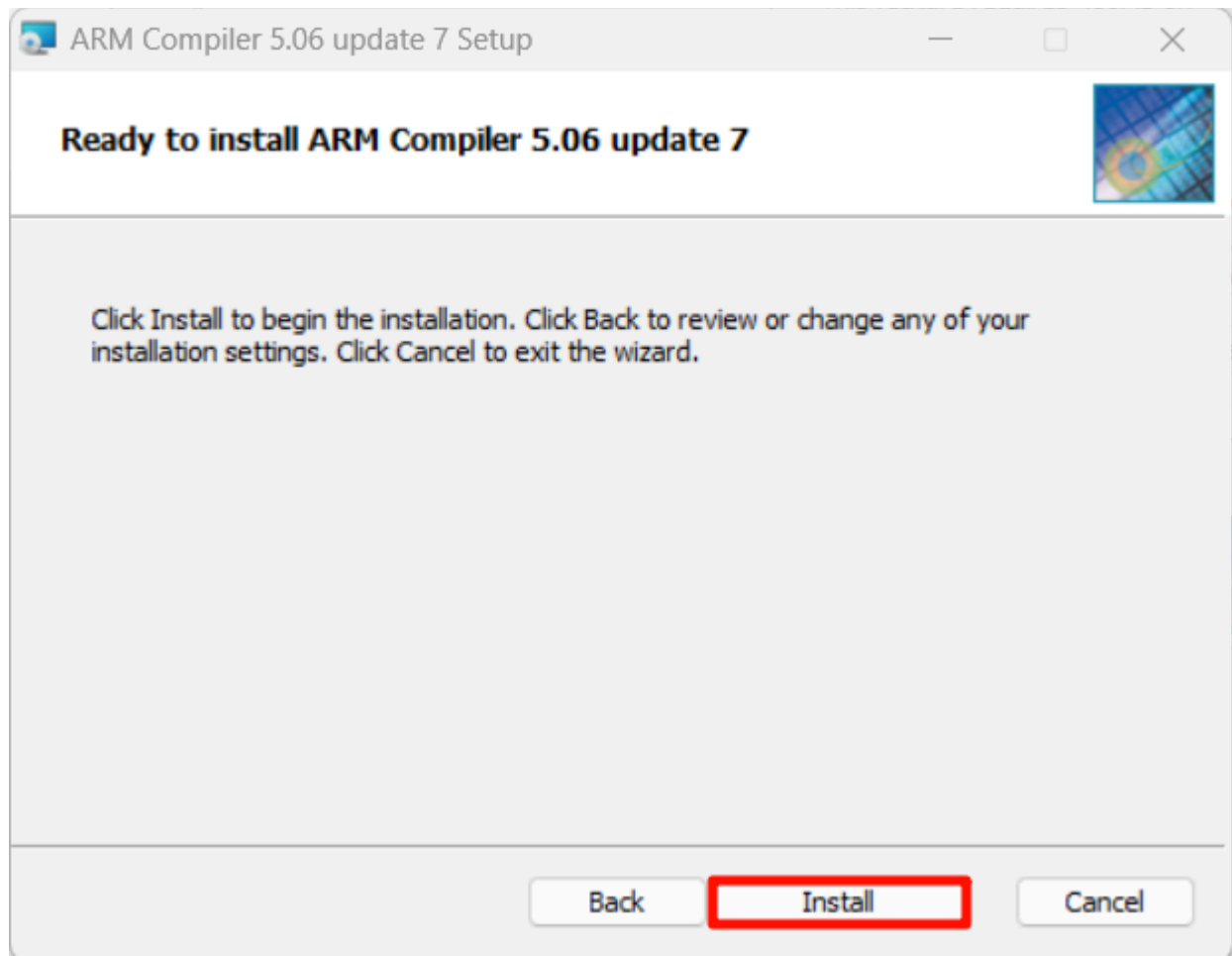


Installation Location

Modify the installation location: Create a new ARMCC folder under C:\Users\W_ML\AppData\Local\Keil_v5\ARM, and select this folder as the AC5 installation location.

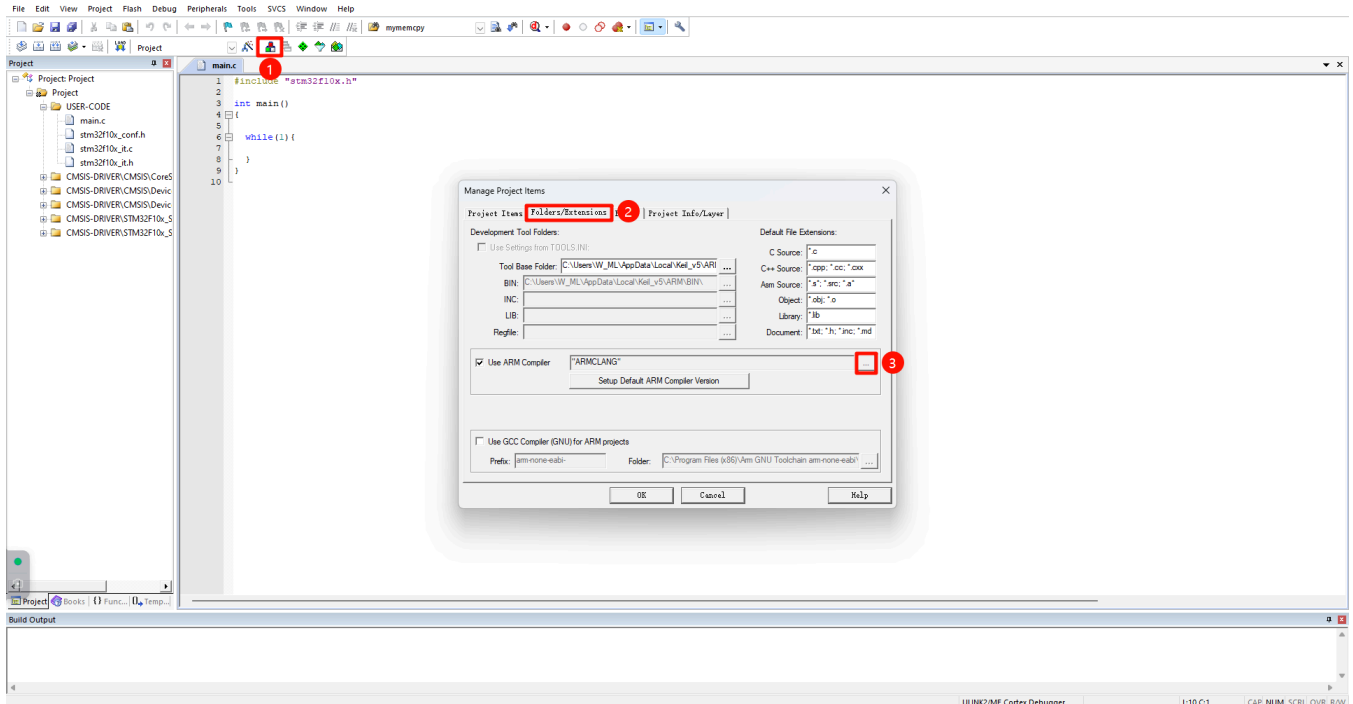
You need to create the ARMCC folder based on your MDK-ARM installation location.



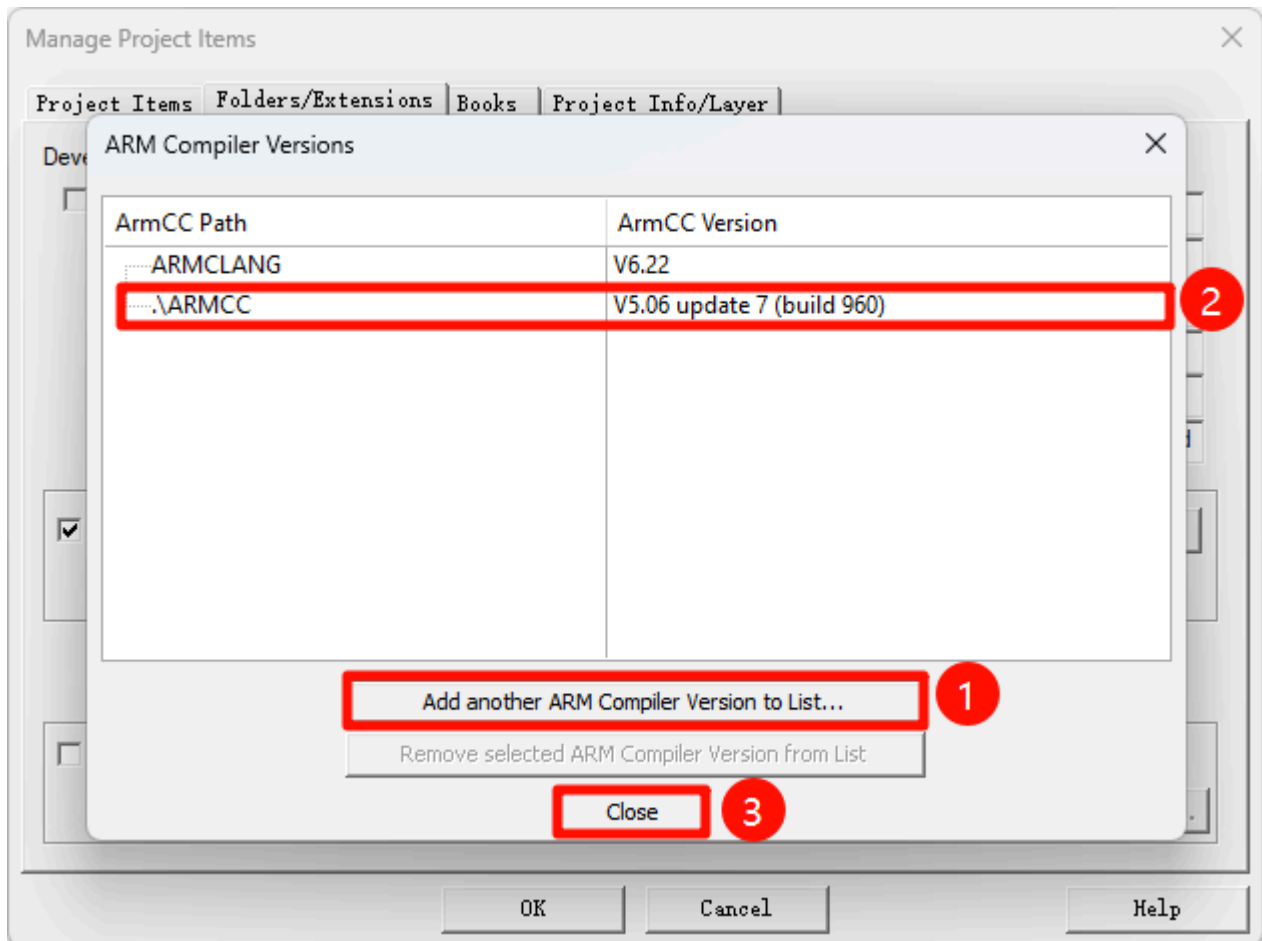


Add AC 5

The following involves operations in MDK-ARM software. You need to open a project; otherwise, the relevant menu options cannot be selected.

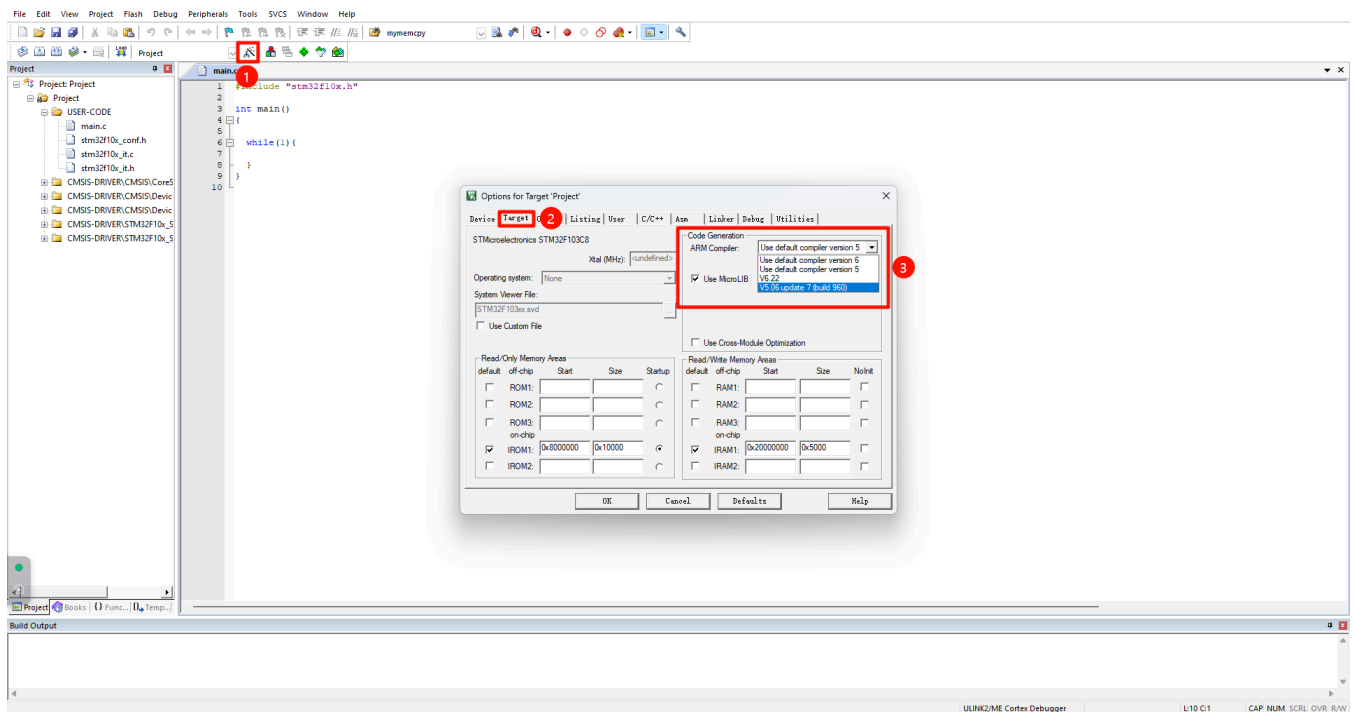


Add compiler: Select C:\Users\W_ML\AppData\Local\Keil_v5\ARM\ARMCC folder



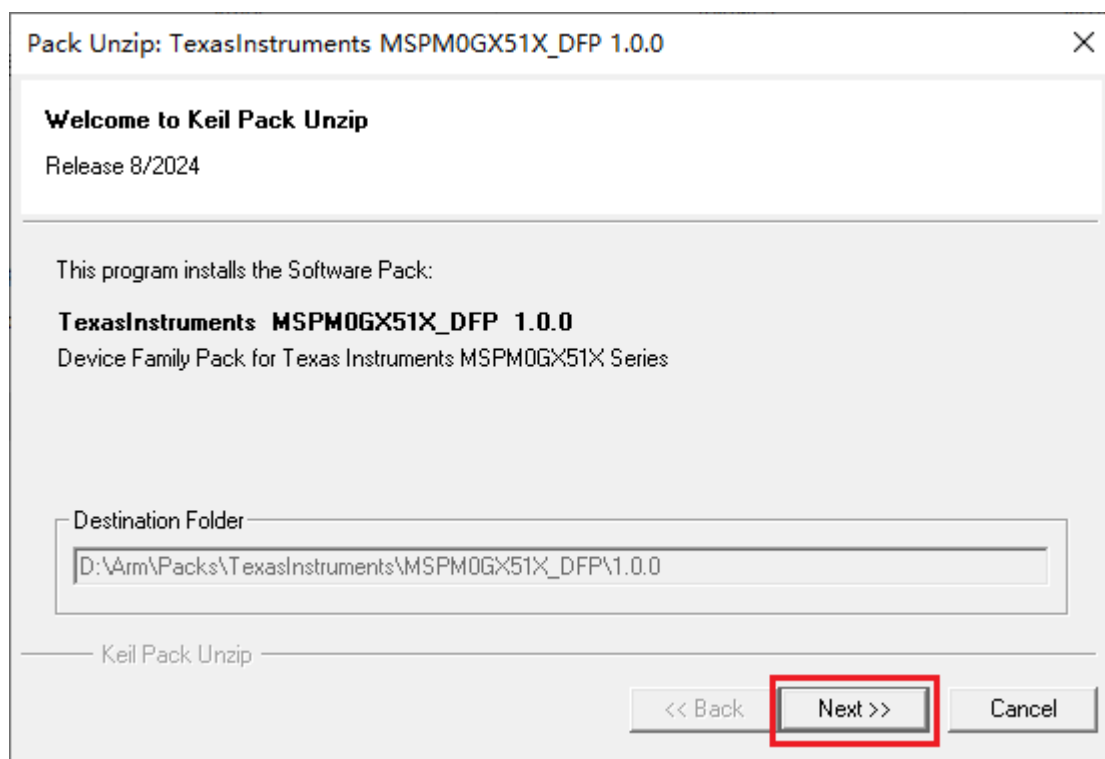
Verify AC 5

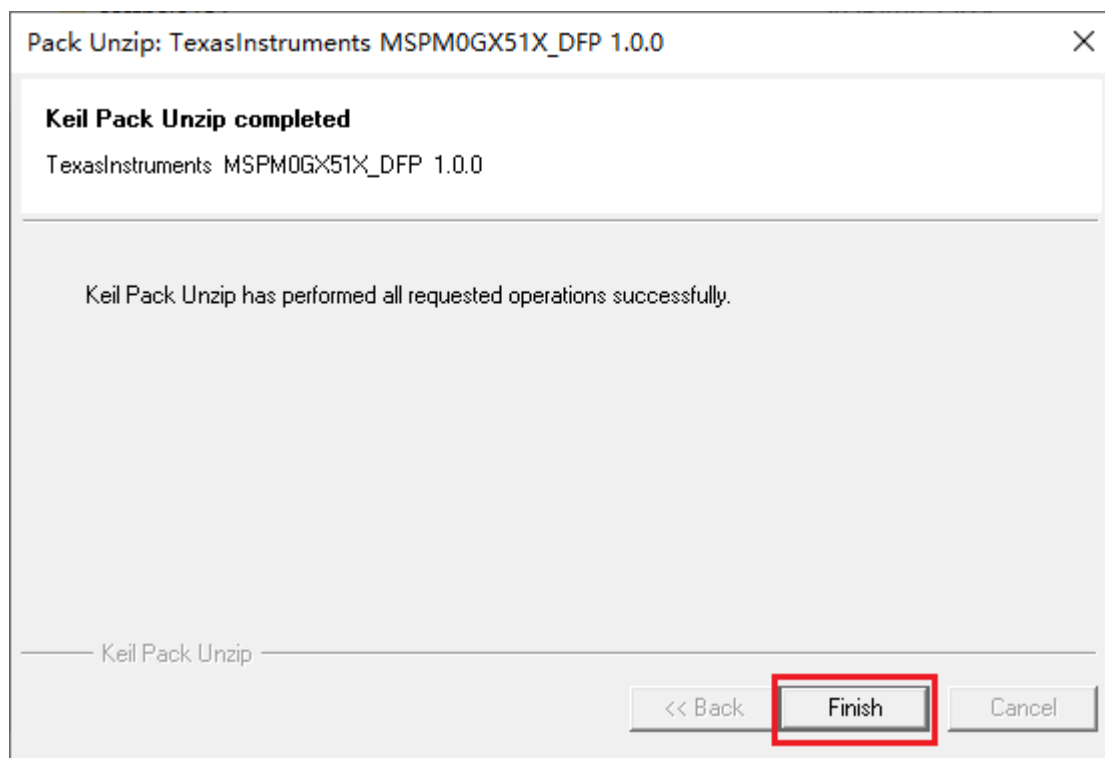
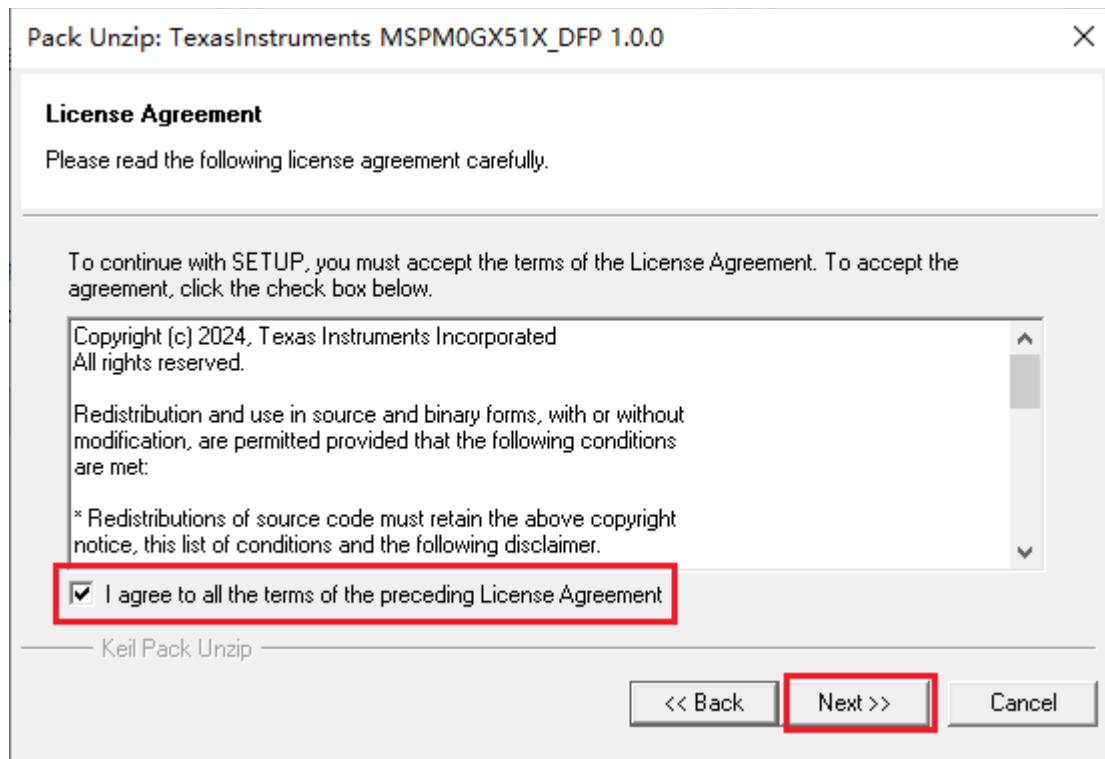
If `compiler version 5` appears, the installation was successful!



Install MSPM0G Chip Pack

Double-click `TexasInstruments.MSPM0GX51X_DFP.1.0.0.pack` to install the chip pack:

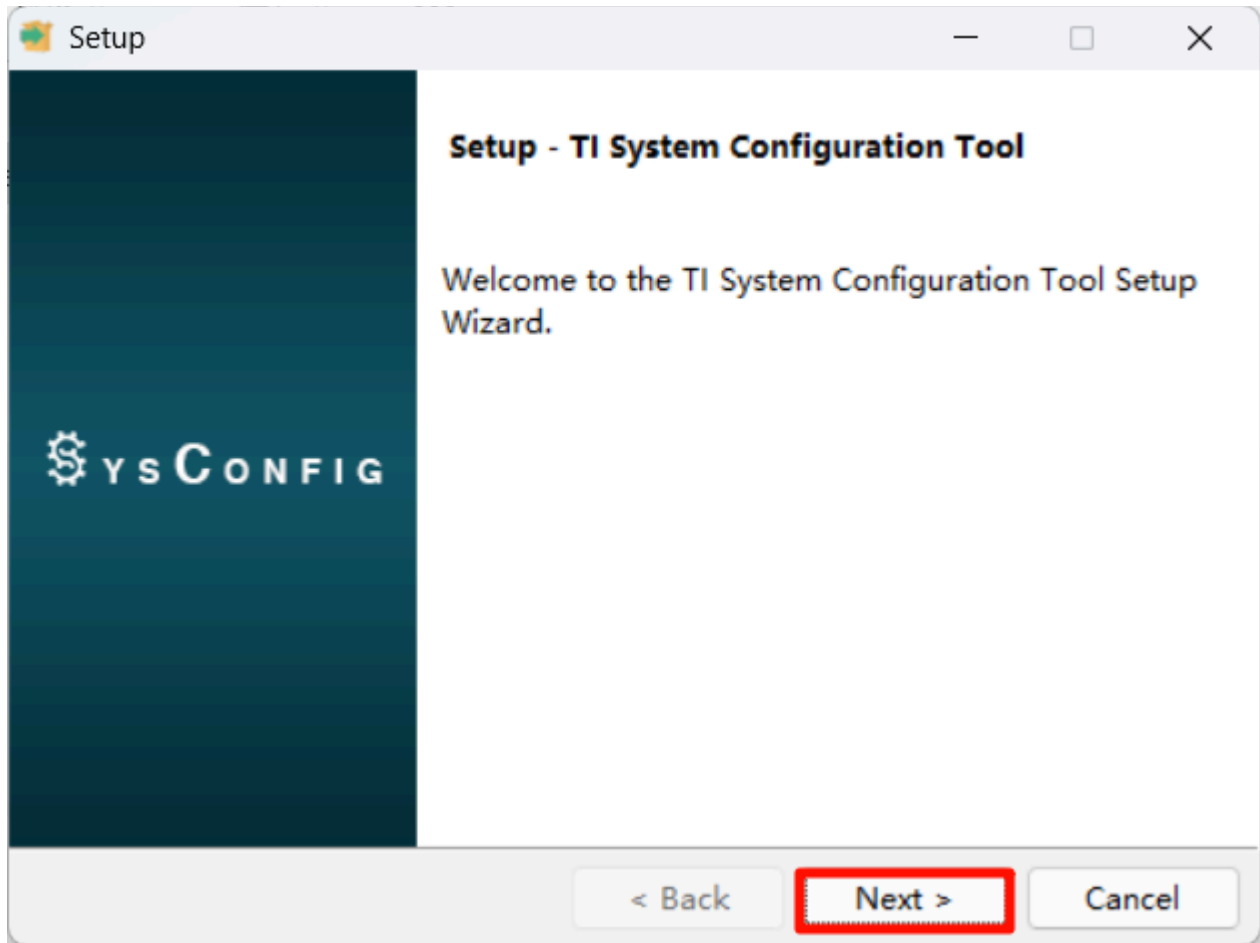




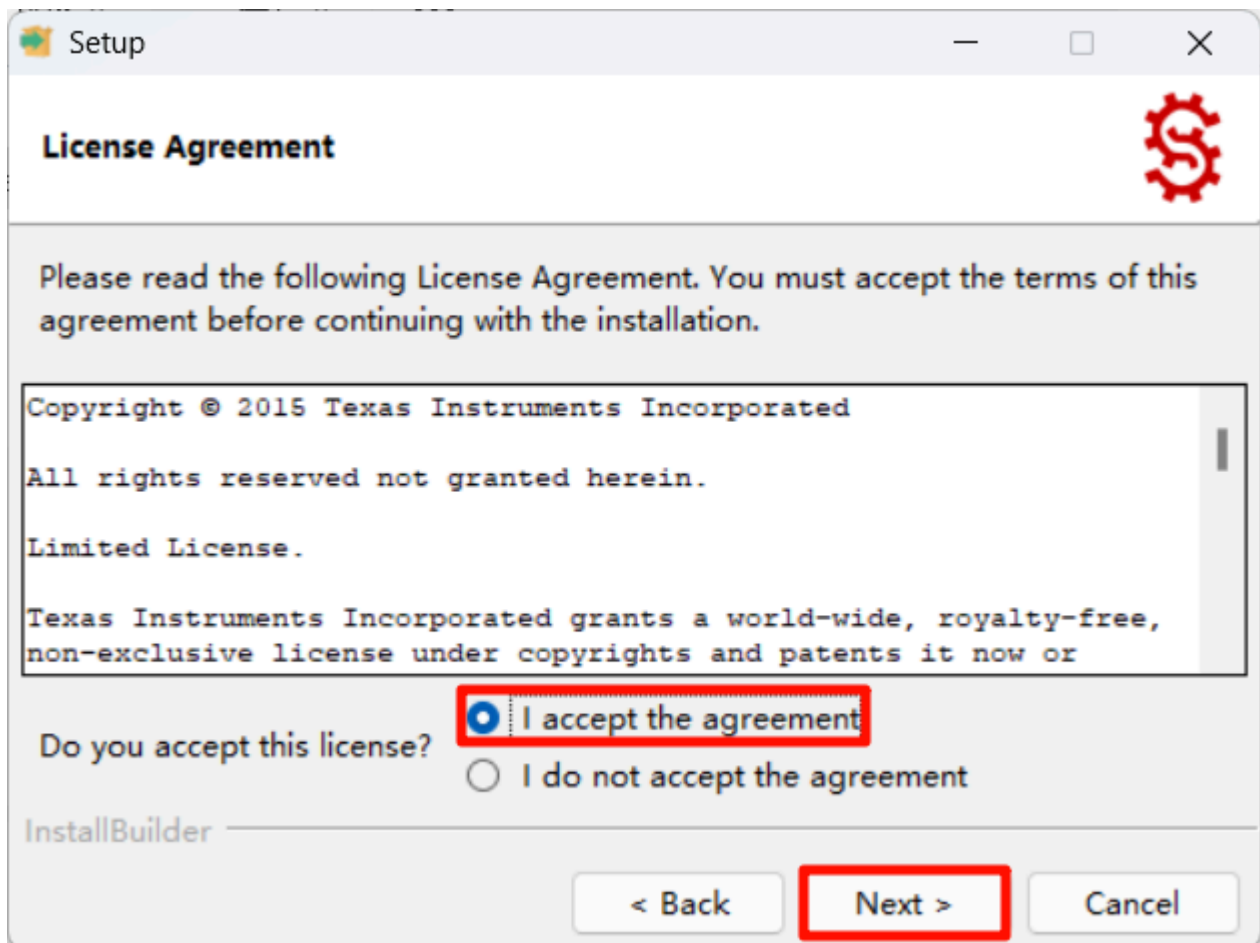
2. SysConfig Installation

SysConfig is a configuration tool for TI microcontrollers and processors, designed to simplify TI developers' hardware and software configuration and accelerate software development!

Open `sysconfig-xxx-setup.exe` as administrator and follow the prompts:

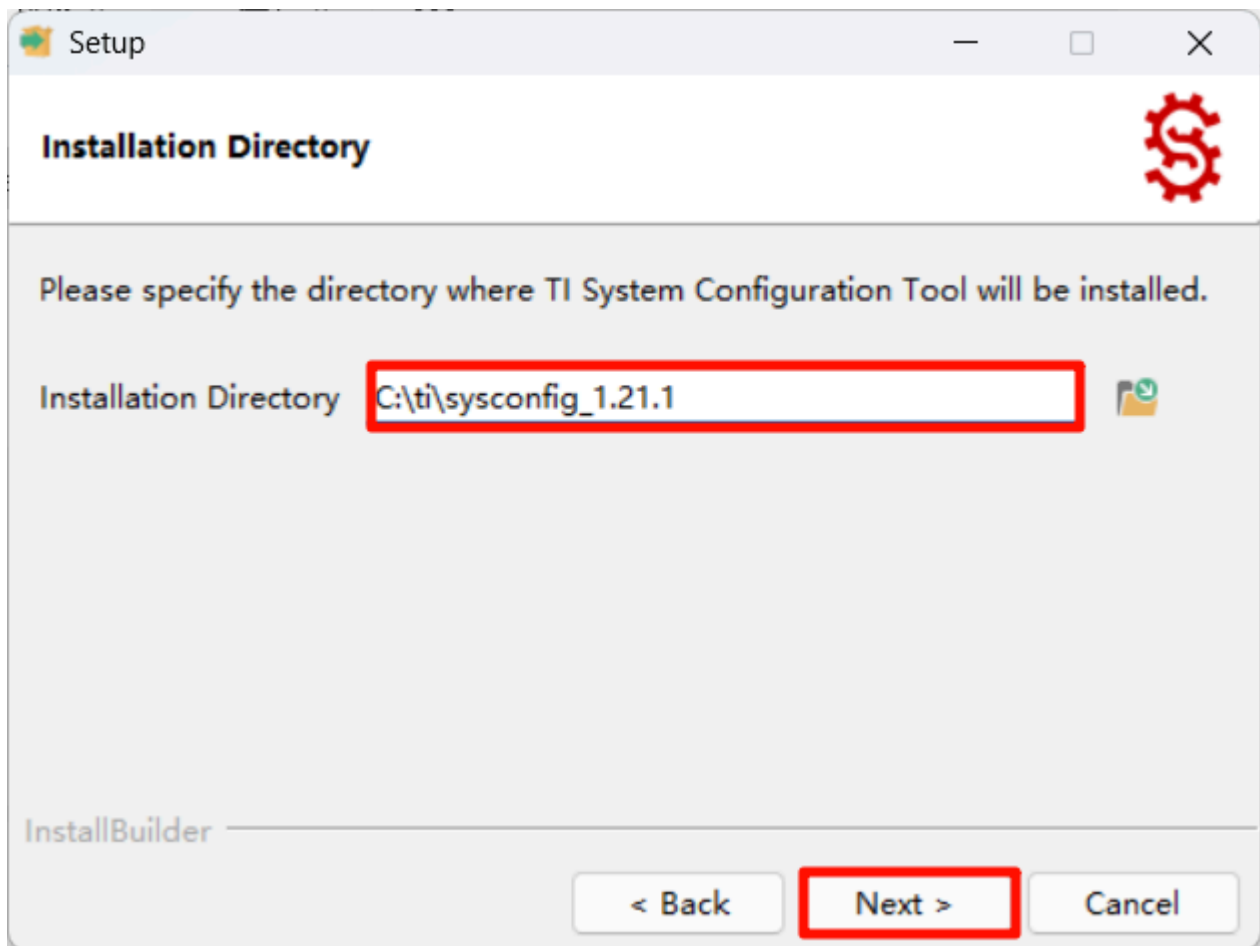


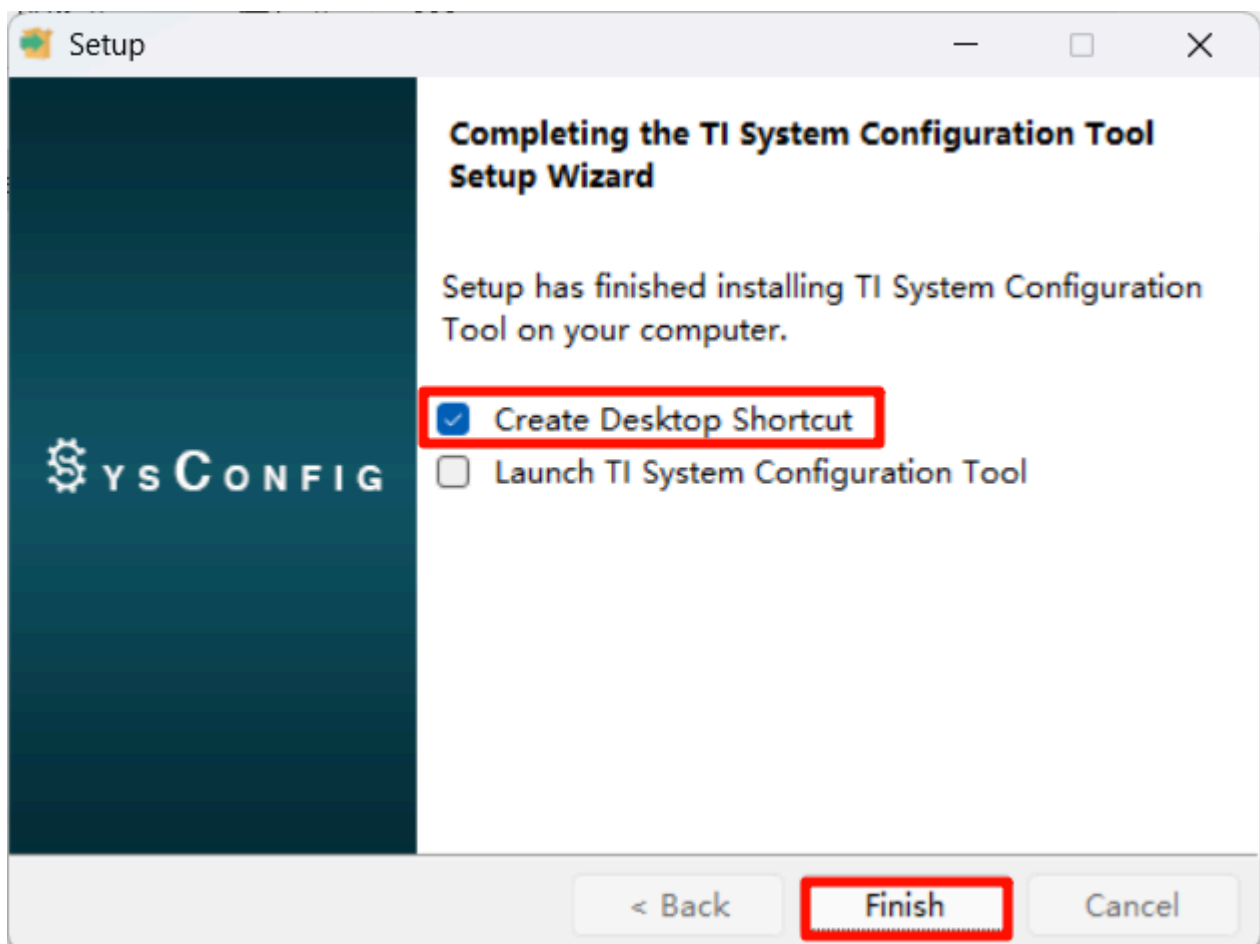
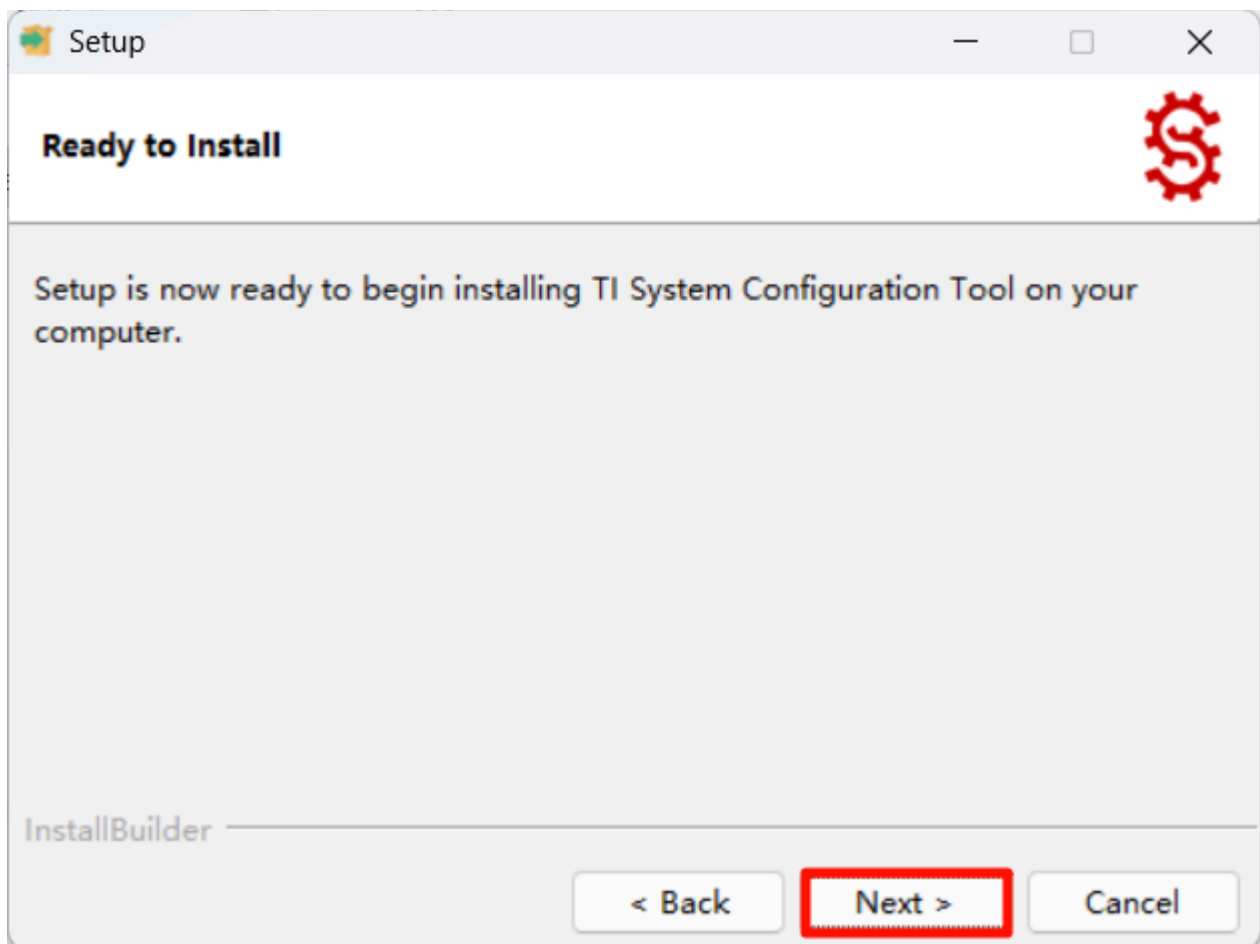
Accept Agreement



Installation Location

It is recommended to use the default location:





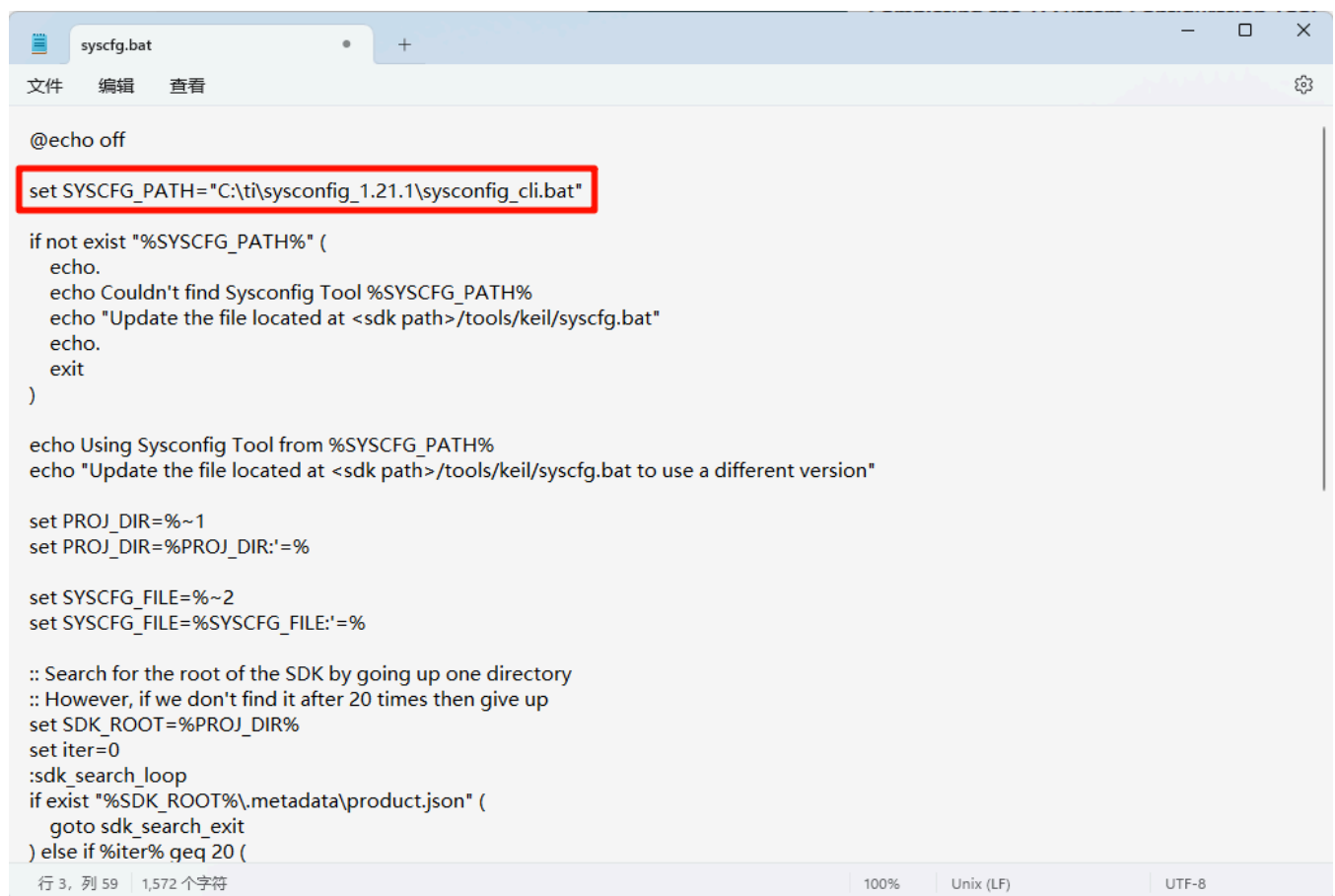
3. Configure SysConfig in KEIL 5

Configure the SysConfig tool for use in KEIL MDK-ARM.

syscfg.bat

Use Notepad to edit the `syscfg.bat` file, then check if `SYSCFG_PATH` is correct: **Fill in the path to the `sysconfig_cli.bat` file under the SysConfig installation directory. Note the version number and path must match!**

```
syscfg.bat file path: C:\ti\mspm0_sdk_2_02_00_05\tools\keil
sysconfig_cli.bat file path: C:\ti\sysconfig_1.21.1
If the user changes the default software installation location, they need to find the
syscfg.bat file under the MSPM0-SDK installation location and modify it themselves.
```



```
@echo off

set SYSCFG_PATH="C:\ti\sysconfig_1.21.1\sysconfig_cli.bat"

if not exist "%SYSCFG_PATH%" (
    echo.
    echo Couldn't find Sysconfig Tool %SYSCFG_PATH%
    echo "Update the file located at <sdk path>/tools/keil/syscfg.bat"
    echo.
    exit
)

echo Using Sysconfig Tool from %SYSCFG_PATH%
echo "Update the file located at <sdk path>/tools/keil/syscfg.bat to use a different version"

set PROJ_DIR=%~1
set PROJ_DIR=%PROJ_DIR:='%%

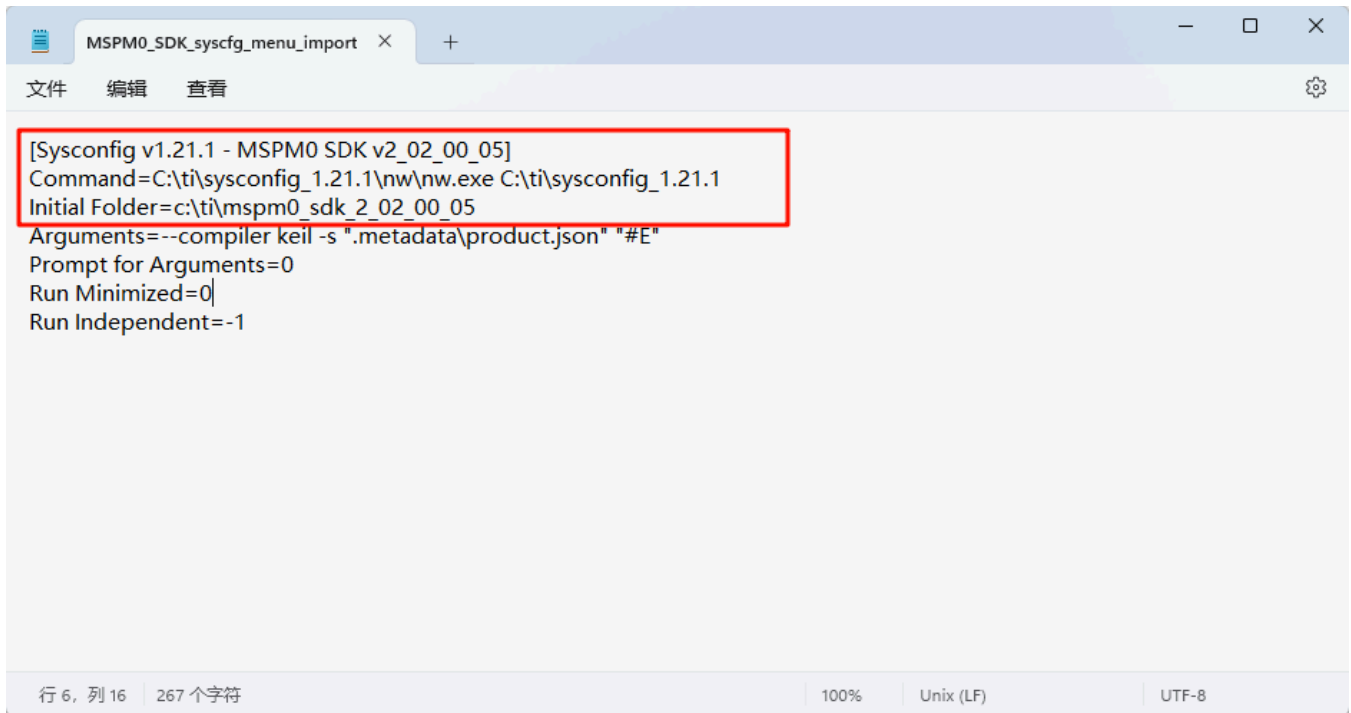
set SYSCFG_FILE=%~2
set SYSCFG_FILE=%SYSCFG_FILE:='%%

:: Search for the root of the SDK by going up one directory
:: However, if we don't find it after 20 times then give up
set SDK_ROOT=%PROJ_DIR%
set iter=0
: sdk_search_loop
if exist "%SDK_ROOT%\metadata\product.json" (
    goto sdk_search_exit
) else if %iter% geq 20 (
    goto sdk_search_exit
) else (
    set SDK_ROOT=%~1\..
    set iter=%iter%+1
    goto sdk_search_loop
)

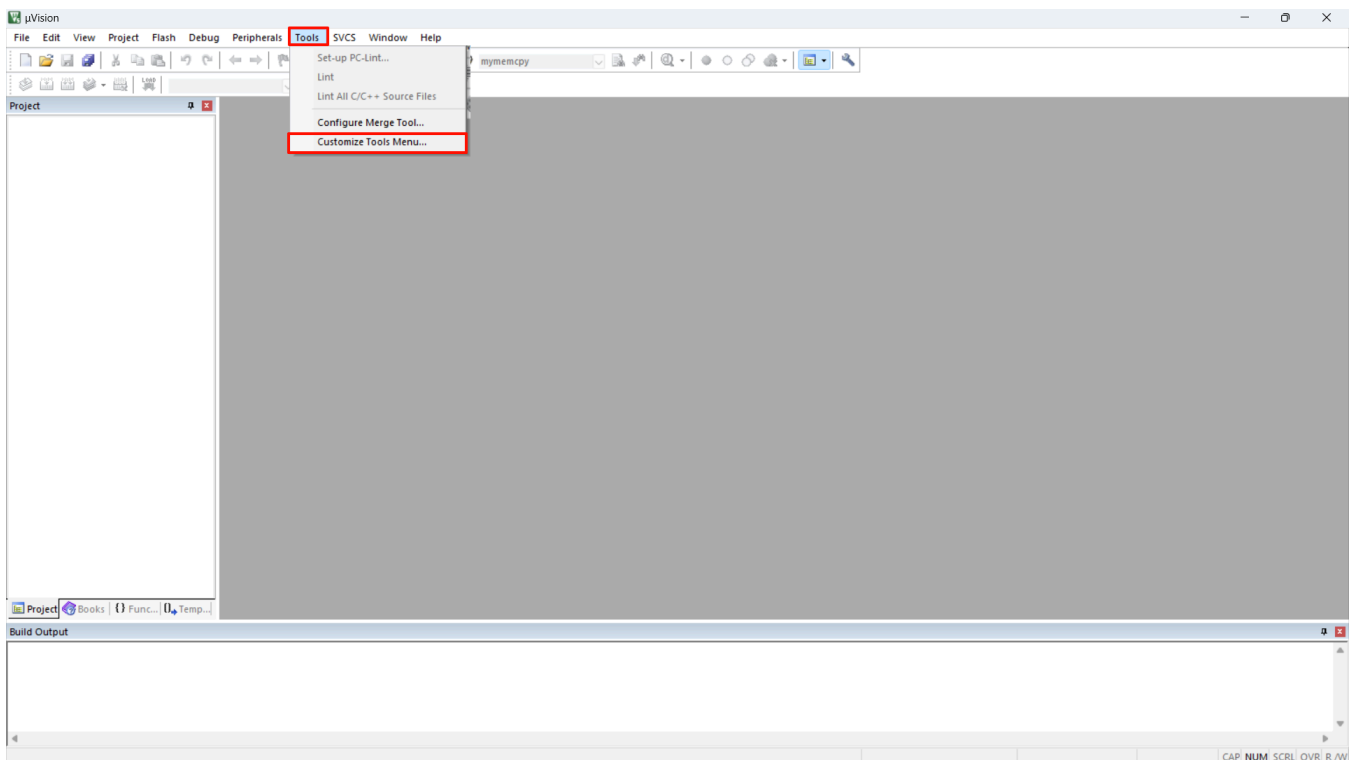
: sdk_search_exit
set SDK_ROOT=%SDK_ROOT%
```

MSPM0_SDK_syscfg_menu_import.cfg

Use Notepad to edit the `MSPM0_SDK_syscfg_menu_import.cfg` file, then check if the path and version number are correct:

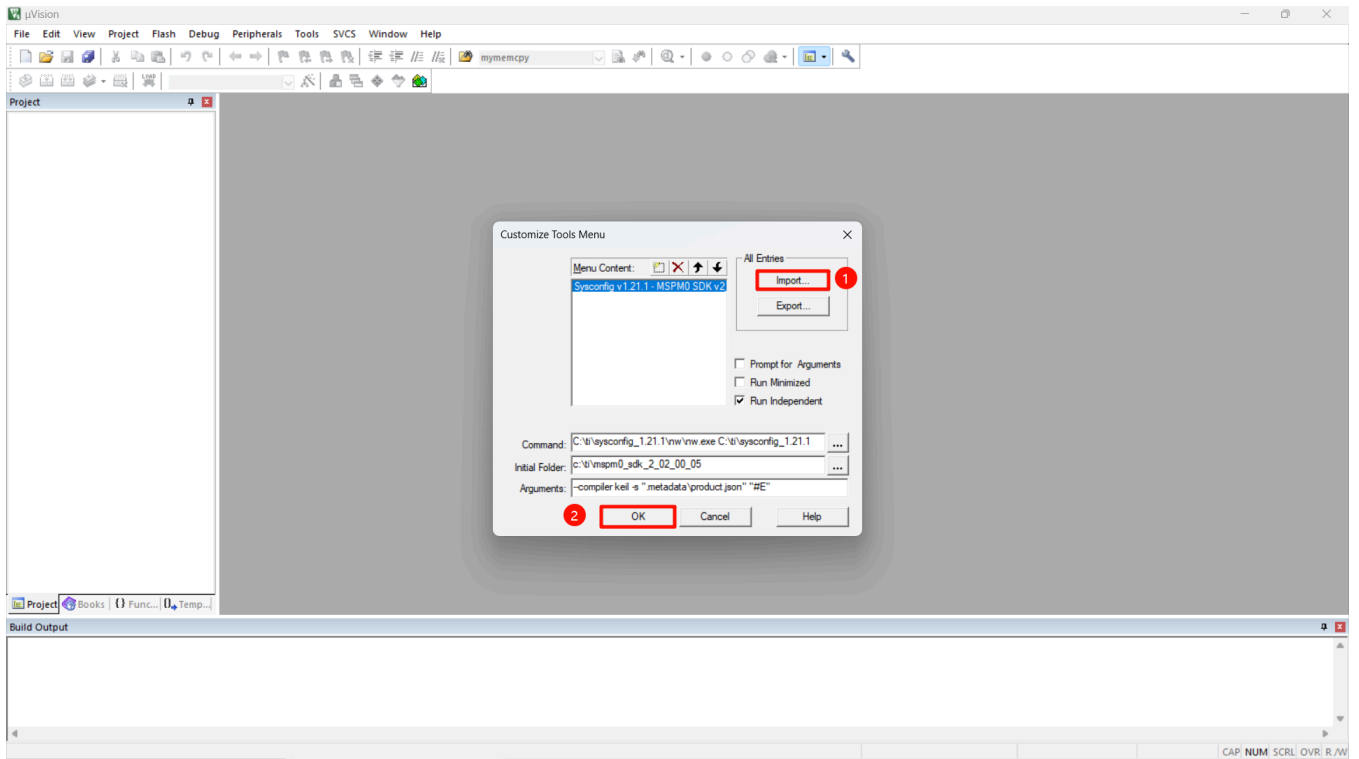


Add SysConfig Tool in KEIL 5

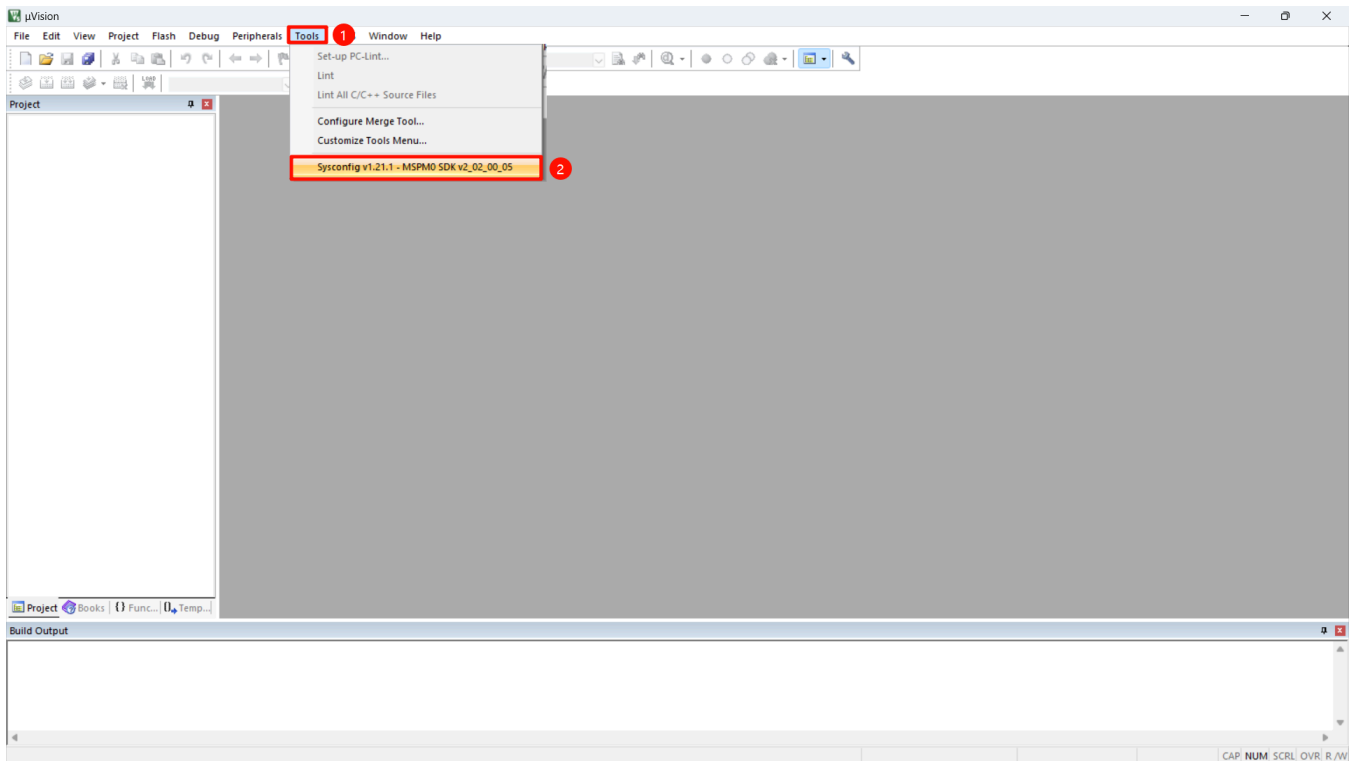


Find and open the `MSPM0_SDK_syscfg_menu_import.cfg` file:

MSPM0_SDK_syscfg_menu_import.cfg file location: `C:\ti\mspm0_sdk_2_02_00_05\tools\keil`
If the user changes the default software installation location, they need to find the `MSPM0_SDK_syscfg_menu_import.bat` file under the MSPM0-SDK installation location themselves.



Use SysConfig Tool



If the following interface appears, the configuration was successful!

opening the sysconfig tool from the .cfg file in TI projects will not show this prompt!

